Understanding the Threat Landscape of Remote Cloud Storage

ECE239AS

Slides created by Katherine Izhikevich

“Origins” of a Data Breach

Zero Day

Phishing Attack

Stolen Credential

“Origins” of a Data Breach

Zero Day

Phishing Attack

Stolen Credential

Unintentional Exposure

- Using a resource that seems to work as intended, but is vulnerable in practice.
- Think: using `1234` as a password.

Data leaks happen all the time

Aston Villa Football Club leaks members' data

21 May 2024 Chris Price Cybersecurity

Aston Villa Football Club (AVFC) left a publicly leaking Amazon Web Services (AWS) S3 bucket containing the personally identifiable information of 135,770 individuals. The affected fans are vulnerable to spear phishing, social engineering attacks, and identity theft attempts.

Data leaks happen all the time

India's national logistics portal exposed sensitive personal data, trade records

Jagmeet Singh / 4:45 AM PDT • October 2, 2023

"... Exposed data included full names, nationality, date of birth, gender, passport numbers, ..."

Data leaks happen all the time

Lantum S3 bucket leak is prescription for chaos for thousands of UK doctors

Lindsay Clark Mon 12 Jun 2023 // 12:34 UTC

"... exposed personal details relating to 3,200 individuals via unsecured S3 buckets… including passport details… medical documents …

Data leaks happen all the time

A Cyberattack Illuminates the Shaky State of Student Privacy

At a moment when education technology firms are stockpiling sensitive information on millions of school children, safeguards for student data have broken down.

By Natasha Singer

July 31, 2022

6 MIN READ

Data leaks happen all the time

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By Natasha Singer
July 31, 2022

6 MIN READ

Microsoft leaked 2.4TB of data belonging to sensitive customer. Critics are furious

Data includes signed contracts and projects related to critical infrastructure.

DAN GOODIN - 10/20/2022, 4:03 PM

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A huge data leak of 1 billion records exposes China's vast surveillance state

One billion resident records were allegedly siphoned from a police database

Zack Whittaker, Carly Page / 1:15 PM EDT • July 7, 2022

Comment

Data leaks happen all the time

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Zack Whittaker, Carly Page / 1:15 PM EDT • July 7, 2022

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Insecure Amazon S3 bucket exposed personal data on 500,000 Ghanaian graduates

John Leyden 06 January 2022 at 10:58 UTC
Updated: 10 January 2022 at 09:40 UTC

Data leaks happen all the time

A Cy ack Illuminates the Shaky State vacy

At a m chnology firms are stock ions of school children, safeguards for stude

By Na

July 3

6 MIN READ

Micros a belonging to ser ious

Sennheiser exposed personal data of 28,000 customers with leaky S3 bucket

Server containing full names, email addresses, phone numbers, and supplier information was left open to the public for three years

by: Danny Bradbury 16 Dec 2021

et exposed anaian

Insecure Am personal data o graduates

John Leyden 06 January 2022 at 10:58 UTC
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A a leak of 1 billion recor es C vast surveillance state

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Data leaks happen all the time

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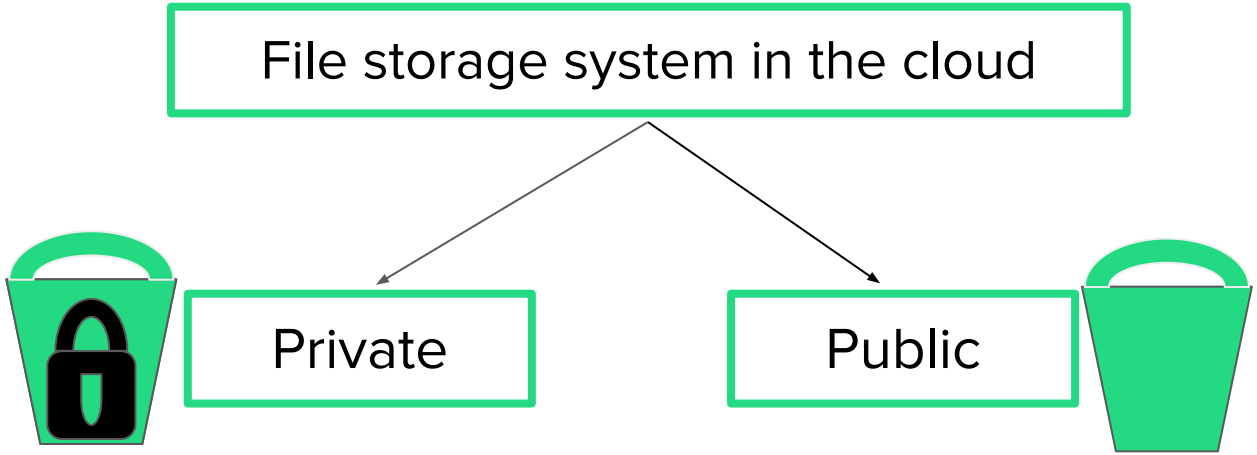
What is a bucket?

File storage system in the cloud

What is a bucket?



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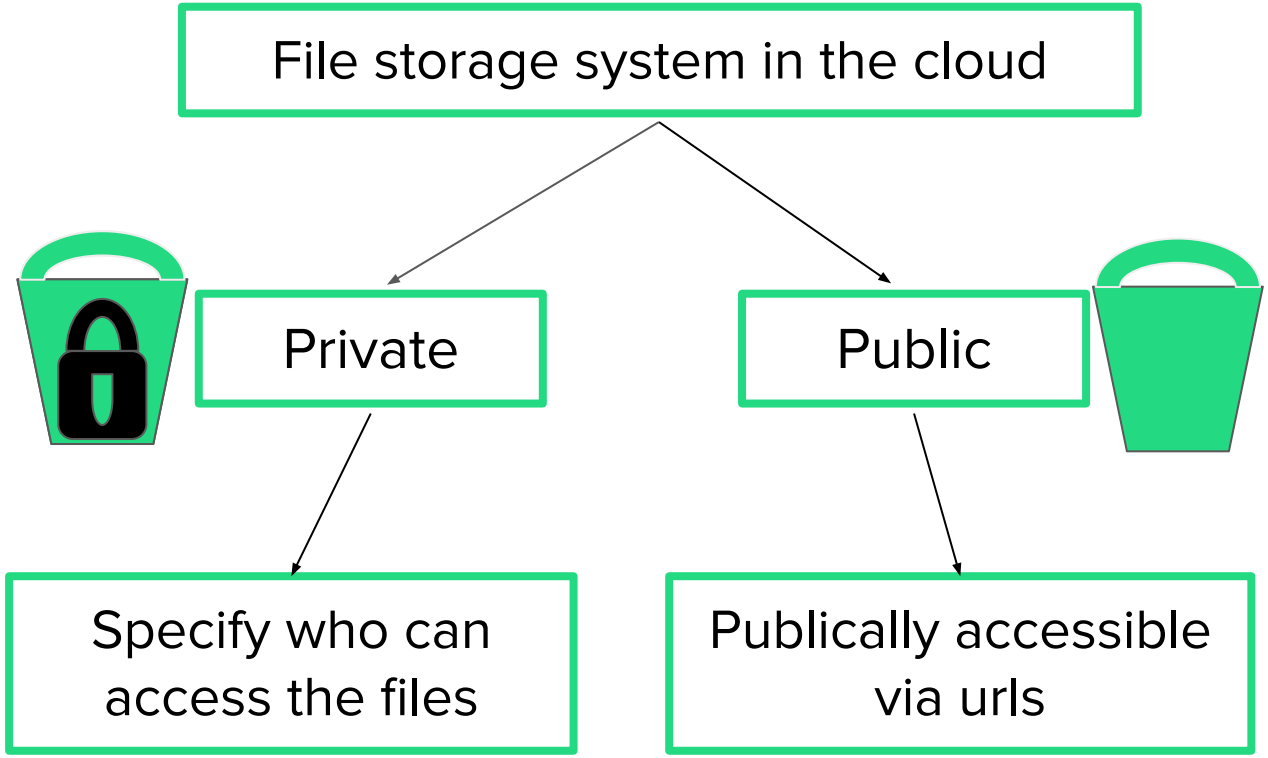


Private buckets store sensitive information

File storage system in the cloud

Private

Specify who can access the files

Storing sensitive information about customers

Private buckets store sensitive information

File storage system in the cloud

Private

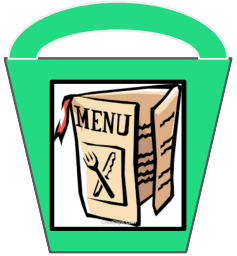
Specify who can access the files

Employees, contractors

Storing sensitive information about customers

Public buckets host public information

File storage system in the cloud



Public

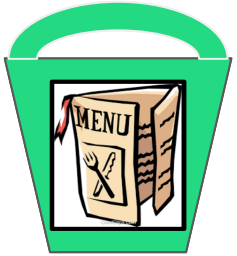
Publically accessible via urls

Public buckets host public information

File storage system in the cloud

Public

Publically accessible via urls



https://**myrestaurant**.s3.us-west-2.amazonaws.com

Public buckets can be configured as a website

https://**mywebsite**.s3.us-west-2.amazonaws.com

Publically accessible
via this url

Creative Freedom to Name Buckets

- Alphanumeric, lowercase only, dashes allowed
- Length 3-63
- Must be an unused/unique name - no repository exists

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  - 3140ebu3b
  - you-get-the-point

Creative Freedom to Name Buckets

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- Examples of valid bucket names (that are available)!
  - thebigtoe
  - 3140ebu3b
  - you-get-the-point
- Astronomical number of possible bucket names: ~10^{101}

Cloud Providers Offering Bucket Service

Cloud Providers Offering Bucket Service

Simple Storage Service

Bucket Recap

File storage system in the cloud with a unique name

Private

Public

Specify who can access the files

Publically accessible via urls

Attackers exploit public buckets meant to be private

File storage system in the cloud with a unique name

Private

Public

Specify who can access the files

Publically accessible via urls

Researchers explore “leaky” cloud infrastructure

ACSAC ‘18

There’s a Hole in that Bucket!
A Large-scale Analysis of Misconfigured S3 Buckets

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Seminal study, guessing bucket names, finding exposed data in s3

RAID ‘21

Stratosphere: Finding Vulnerable Cloud Storage Buckets

Jack Cable*
Stanford University

Drew Gregory*
Stanford University

Liz Izhikevich*
Stanford University

Zakir Durumeric
Stanford University

Faster, ML-based bucket name generator across s3, GCP, and Alibaba

EuroS&P ‘24

Using Honeybuckets to Characterize Cloud Storage Scanning in the Wild

Katherine Izhikevich
UC San Diego

Geoffrey M. Voelker
UC San Diego

Stefan Savage
UC San Diego

Liz Izhikevich
Stanford University

Deploying honeypots to understand threat landscape of exposures

The Bucket Threat Model

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Threat Model

Readable Buckets	- *Data leakage*
Writable Buckets	- *Ransom Demand* - *Web Resource Infection*
Unclaimed Buckets	- *Dangling Subdomain Takeover*

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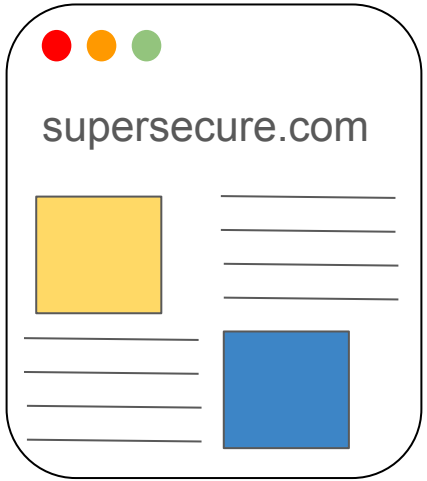

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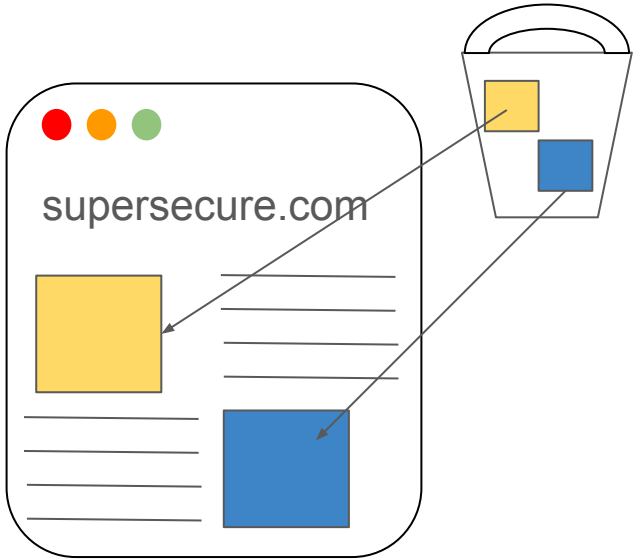

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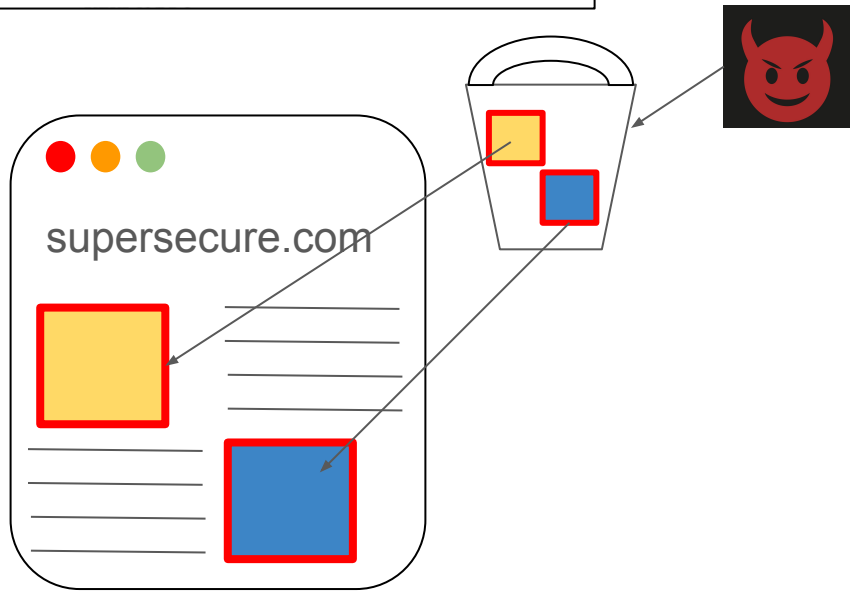

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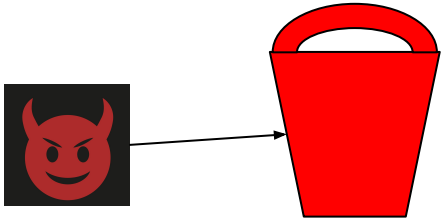
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